

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

A Rugged and Compact Time-of-Flight Mass Spectrometer for Fast and Sensitive Leak Detection

Authors: Marc Gonin, Katrin Fuhrer, Michael Ugarov, Val Vaughn, Steve Ulrich, Michael McCully, Albert Schultz

Institute: Ionwerks, Inc.

ABSTRACT

A compact Time-of-Flight mass spectrometer is being developed for air leak detection during the space shuttle start. The main challenges for this application are speed, sensitivity, dynamic range, and ruggedness. We made several approaches to address these issues for a compact TOF prototype. A rugged, filament-less ionizer has been developed, based on a surface desorption process. Advantages and current limitation of this ionizer are discussed. The sensitivity of a TOF is principally limited by the saturation of the Micro-channel-plate detector, which can handle only a few million counts per second. An approach to overcome this limitation for this application is discussed. Further a detector scheme is presented which allows to obtain a dynamic range of 10^6 per second with a time-to-digital converter.